

Homework 2: Solutions

STT 861 — Theory of Probability and Statistics I

Casella and Berger, *Statistical Inference*, Chapter 1, pp. 40–44

Homework problems: 1–12, corresponding to Exercises 1.24, 1.26, 1.27, 1.34, 1.38, 1.42, 1.46, 1.47, 1.49, 1.51, 1.53, and 1.55.

Problem 1

C&B Exercise 1.24, p. 40

Players A and B flip a coin alternately and independently, with A flipping first. The first player to obtain a head wins. Find A 's winning probability for a fair coin and for a coin with head probability p , and show that this probability exceeds $1/2$ when $0 < p < 1$.

Solution

Player A flips on tosses $1, 3, 5, \dots$. For $k = 0, 1, 2, \dots$, A wins on toss $2k + 1$ precisely when the first $2k$ tosses are tails and the next toss is a head. These winning events are disjoint.

(a) **Fair coin.** Independence and the geometric-series formula give

$$\mathbb{P}(A \text{ wins}) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{2k+1} = \frac{1/2}{1 - 1/4} = \boxed{\frac{2}{3}}.$$

(b) **Head probability p .** For $0 < p \leq 1$,

$$\mathbb{P}(A \text{ wins}) = \sum_{k=0}^{\infty} (1-p)^{2k} p = \frac{p}{1 - (1-p)^2} = \boxed{\frac{1}{2-p}}.$$

The game terminates with probability one, since the probability of no head in the first m tosses is $(1-p)^m \rightarrow 0$. If $p = 0$, neither player ever wins, so $\mathbb{P}(A \text{ wins}) = 0$; the simplified formula does not apply at that endpoint.

(c) **Advantage of going first.** When $0 < p < 1$,

$$\mathbb{P}(A \text{ wins}) - \frac{1}{2} = \frac{1}{2-p} - \frac{1}{2} = \frac{p}{2(2-p)} > 0.$$

Thus A has a strictly greater than $1/2$ chance of winning.

Problem 2

C&B Exercise 1.26, p. 40

A fair die is rolled independently until the first 6 appears. Find the probability that more than five rolls are required.

Solution

Let T be the roll on which the first 6 occurs. The event $T > 5$ is exactly the event that none of the first five rolls is a 6. Therefore

$$\mathbb{P}(T > 5) = \left(\frac{5}{6}\right)^5 = \frac{3125}{7776} \approx 0.4019.$$

The sixth roll is unrestricted; it need not be the first 6.

Problem 3**C&B Exercise 1.27, p. 40**

Verify the following binomial identities for $n \geq 2$:

$$(a) \sum_{k=0}^n (-1)^k \binom{n}{k} = 0, \quad (b) \sum_{k=1}^n k \binom{n}{k} = n2^{n-1}, \quad (c) \sum_{k=1}^n (-1)^{k+1} k \binom{n}{k} = 0.$$

Solution

(a) By the binomial theorem,

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = (1-1)^n = \boxed{0}.$$

(b) Differentiate the polynomial identity $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$ to obtain

$$n(1+x)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^{k-1}.$$

Setting $x = 1$ gives

$$\boxed{\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}}.$$

(c) Instead set $x = -1$ in the differentiated identity. Because $n-1 \geq 1$ and $(-1)^{k-1} = (-1)^{k+1}$,

$$\boxed{\sum_{k=1}^n (-1)^{k+1} k \binom{n}{k} = n(1-1)^{n-1} = 0}.$$

Problem 4**C&B Exercise 1.34, p. 41**

Litter 1 contains two brown-haired and one gray-haired offspring; litter 2 contains three brown-haired and two gray-haired offspring. Choose a litter with equal probability and then choose an offspring uniformly from that litter. Find the probability of selecting a brown-haired offspring and, given that it is brown-haired, the probability that it came from litter 1.

Solution

Let L_i denote selection of litter i , and let B denote a brown-haired offspring. Then

$$\mathbb{P}(L_1) = \mathbb{P}(L_2) = \frac{1}{2}, \quad \mathbb{P}(B | L_1) = \frac{2}{3}, \quad \mathbb{P}(B | L_2) = \frac{3}{5}.$$

(a) The law of total probability gives

$$\boxed{\mathbb{P}(B) = \frac{2}{3} \cdot \frac{1}{2} + \frac{3}{5} \cdot \frac{1}{2} = \frac{19}{30}}.$$

(b) By Bayes' theorem,

$$\boxed{\mathbb{P}(L_1 | B) = \frac{\mathbb{P}(B | L_1) \mathbb{P}(L_1)}{\mathbb{P}(B)} = \frac{(2/3)(1/2)}{19/30} = \frac{10}{19}}.$$

Problem 5**C&B Exercise 1.38, p. 42**

Prove the four conditional-probability identities below, assuming that every conditioning event has positive probability.

Solution

(a) If $\mathbb{P}(B) = 1$, then $\mathbb{P}(A | B) = \mathbb{P}(A)$. Since $A \cap B^c \subseteq B^c$ and $\mathbb{P}(B^c) = 0$, we have $\mathbb{P}(A \cap B^c) = 0$. The disjoint decomposition $A = (A \cap B) \cup (A \cap B^c)$ implies $\mathbb{P}(A \cap B) = \mathbb{P}(A)$. Hence

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A).$$

(b) If $A \subseteq B$, then $\mathbb{P}(B | A) = 1$ and $\mathbb{P}(A | B) = \mathbb{P}(A)/\mathbb{P}(B)$. Because $A \cap B = A$,

$$\mathbb{P}(B | A) = \frac{\mathbb{P}(A)}{\mathbb{P}(A)} = 1, \quad \mathbb{P}(A | B) = \frac{\mathbb{P}(A)}{\mathbb{P}(B)}.$$

(c) If $A \cap B = \emptyset$, then $\mathbb{P}(A | A \cup B) = \mathbb{P}(A)/(\mathbb{P}(A) + \mathbb{P}(B))$. Here $A \cap (A \cup B) = A$, and disjointness gives $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$. Therefore

$$\mathbb{P}(A | A \cup B) = \frac{\mathbb{P}(A \cap (A \cup B))}{\mathbb{P}(A \cup B)} = \frac{\mathbb{P}(A)}{\mathbb{P}(A) + \mathbb{P}(B)}.$$

(d) **Multiplication rule for three events.** Using $\mathbb{P}(B \cap C) > 0$ and $\mathbb{P}(C) > 0$,

$$\begin{aligned} \mathbb{P}(A | B \cap C) \mathbb{P}(B | C) \mathbb{P}(C) &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(B \cap C)} \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} \mathbb{P}(C) \\ &= \mathbb{P}(A \cap B \cap C). \end{aligned}$$

Problem 6**C&B Exercise 1.42, p. 43**

Prove the inclusion–exclusion identity by counting how often outcomes belonging to exactly k events are included in the intersection sums. For $1 \leq j \leq n$, define

$$P_j = \sum_{1 \leq i_1 < \dots < i_j \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_j}),$$

and let E_k be the event that exactly k of $A_1, \dots, A_n$ occur.

Solution

(a)

Statement. Show that

$$P \left(\bigcup_{i=1}^n A_i \right) = \sum_{k=1}^n P(E_k).$$

Proof. We first verify that the E_k 's partition $\bigcup_{i=1}^n A_i$.

Step 1: The E_k 's are pairwise disjoint. By definition, a sample point s belongs to exactly one E_k : the unique k equal to the number of the A_i 's that contain s . Hence $E_j \cap E_k = \emptyset$ for $j \neq k$.

Step 2: Their union is $\bigcup_{i=1}^n A_i$. A point s belongs to $\bigcup_{i=1}^n A_i$ if and only if it belongs to at least one A_i , i.e. it belongs to exactly $k \geq 1$ of them for some k . This is precisely the condition $s \in E_k$ for some $k \in \{1, \dots, n\}$. Therefore

$$\bigcup_{i=1}^n A_i = \bigsqcup_{k=1}^n E_k \quad (\text{disjoint union}).$$

Step 3: Apply countable additivity. Since the E_k 's are disjoint and their union equals $\bigcup_{i=1}^n A_i$, by the countable additivity axiom (Axiom 3 of Kolmogorov):

$$P\left(\bigcup_{i=1}^n A_i\right) = P\left(\bigsqcup_{k=1}^n E_k\right) = \sum_{k=1}^n P(E_k).$$

□

(b)

Statement. If E_1 is not empty, show that

$$P(E_1) = \sum_{i=1}^n P(A_i).$$

Proof. A sample point s belongs to E_1 if and only if it belongs to *exactly one* of the events $A_1, \dots, A_n$.

Step 1: Express E_1 as a disjoint union. Define $B_i = A_i \cap E_1$ for each $i = 1, \dots, n$. Since every point of E_1 belongs to exactly one A_i , the sets $B_1, \dots, B_n$ are pairwise disjoint and

$$E_1 = \bigsqcup_{i=1}^n B_i = \bigsqcup_{i=1}^n (A_i \cap E_1).$$

By countable additivity:

$$P(E_1) = \sum_{i=1}^n P(A_i \cap E_1). \quad (*)$$

Step 2: Decompose $P(A_i)$ using the partition $\{E_1, E_2, \dots, E_n\}$. Since $\{E_k\}_{k=1}^n$ partitions $\bigcup_{j=1}^n A_j$ (by part (a)),

$$A_i = \bigsqcup_{k=1}^n (A_i \cap E_k).$$

Hence

$$P(A_i) = \sum_{k=1}^n P(A_i \cap E_k). \quad (**)$$

Step 3: Sum over all i and compare. Summing $(**)$ over $i = 1, \dots, n$:

$$\sum_{i=1}^n P(A_i) = \sum_{i=1}^n \sum_{k=1}^n P(A_i \cap E_k) = \sum_{k=1}^n \sum_{i=1}^n P(A_i \cap E_k). \quad (***)$$

Now fix $k \geq 2$. A point $s \in E_k$ belongs to exactly $k \geq 2$ of the A_i 's, so $\sum_{i=1}^n P(A_i \cap E_k) = k \cdot P(E_k)$. For $k = 1$, each point of E_1 belongs to exactly one A_i , so $\sum_{i=1}^n P(A_i \cap E_1) = P(E_1)$. Therefore $(***)$ becomes:

$$\sum_{i=1}^n P(A_i) = P(E_1) + \sum_{k=2}^n kP(E_k).$$

This does *not* immediately give $\sum P(A_i) = P(E_1)$ in general.

Correction (the statement as given in CB). Part (b) is a restricted statement: it asserts only that

$$P(E_1) = \sum_{i=1}^n P(A_i) \quad \text{when all sample points belong to at most one event,}$$

i.e. when $E_k = \emptyset$ for all $k \geq 2$. In that special case, $A_i \cap E_k = \emptyset$ for $k \geq 2$, so (***) reduces to

$$\sum_{i=1}^n P(A_i) = \sum_{i=1}^n P(A_i \cap E_1) = P(E_1),$$

where the last equality is (*). □

Remark. The more natural reading is: the proof of part (b) establishes the *base contribution* of E_1 to $P_1 = \sum_i P(A_i)$, namely that each point in E_1 is counted exactly once in the sum $\sum_{i=1}^n P(A_i)$. This is the first step in the inclusion-exclusion counting argument that parts (c)–(e) complete. □

(c) Multiplicities in the intersection sums. For parts (c)–(e), return to arbitrary events, allowing overlaps. Write $[n] = \{1, \dots, n\}$. For each $I \subseteq [n]$, define

$$B_I = \left(\bigcap_{i \in I} A_i \right) \cap \left(\bigcap_{i \notin I} A_i^c \right),$$

where an intersection over an empty index set is the sample space. The regions B_I are pairwise disjoint, and B_I consists of the outcomes belonging to precisely the events indexed by I . Thus

$$E_k = \bigsqcup_{\substack{I \subseteq [n] \\ |I|=k}} B_I, \quad \mathbb{P}(E_k) = \sum_{\substack{I \subseteq [n] \\ |I|=k}} \mathbb{P}(B_I).$$

For a fixed $J \subseteq [n]$ with $|J| = j$, an outcome belongs to $\bigcap_{i \in J} A_i$ exactly when its region B_I has $J \subseteq I$. Consequently,

$$\bigcap_{i \in J} A_i = \bigsqcup_{\substack{I \subseteq [n] \\ J \subseteq I}} B_I.$$

Substituting this disjoint decomposition and interchanging the finite sums,

$$\begin{aligned} P_j &= \sum_{\substack{J \subseteq [n] \\ |J|=j}} \mathbb{P} \left(\bigcap_{i \in J} A_i \right) \\ &= \sum_{\substack{J \subseteq [n] \\ |J|=j}} \sum_{\substack{I \subseteq [n] \\ J \subseteq I}} \mathbb{P}(B_I) \\ &= \sum_{\substack{I \subseteq [n] \\ |I| \geq j}} \left(\sum_{\substack{J \subseteq I \\ |J|=j}} 1 \right) \mathbb{P}(B_I) \end{aligned}$$

$$\begin{aligned}
&= \sum_{\substack{I \subseteq [n] \\ |I| \geq j}} \binom{|I|}{j} \mathbb{P}(B_I) \\
&= \sum_{k=j}^n \binom{k}{j} \sum_{\substack{I \subseteq [n] \\ |I|=k}} \mathbb{P}(B_I) \\
&= \boxed{\sum_{k=j}^n \binom{k}{j} \mathbb{P}(E_k)}.
\end{aligned}$$

Here a set I of size k contains exactly $\binom{k}{j}$ subsets J of size j . In particular, the total contribution from the regions making up E_k is

$$\sum_{\substack{I \subseteq [n] \\ |I|=k}} \binom{k}{j} \mathbb{P}(B_I) = \binom{k}{j} \mathbb{P}(E_k).$$

Thus $\mathbb{P}(E_k)$ contributes k times to P_1 , $\binom{k}{2}$ times to P_2 , $\binom{k}{3}$ times to P_3 , and so on.

(d) Alternating multiplicity. For each $k \geq 1$, the binomial theorem gives

$$0 = (1 - 1)^k = 1 + \sum_{j=1}^k (-1)^j \binom{k}{j}.$$

Rearranging,

$$\boxed{\sum_{j=1}^k (-1)^{j+1} \binom{k}{j} = k - \binom{k}{2} + \binom{k}{3} - \cdots + (-1)^{k+1} \binom{k}{k} = 1}.$$

(e) Assemble the identity. Substitute part (c), interchange the finite sums, and use parts (d) and (a):

$$\begin{aligned}
\sum_{j=1}^n (-1)^{j+1} P_j &= \sum_{j=1}^n (-1)^{j+1} \sum_{k=j}^n \binom{k}{j} \mathbb{P}(E_k) \\
&= \sum_{k=1}^n \mathbb{P}(E_k) \left[\sum_{j=1}^k (-1)^{j+1} \binom{k}{j} \right] \\
&= \sum_{k=1}^n \mathbb{P}(E_k) = \boxed{\mathbb{P}\left(\bigcup_{i=1}^n A_i\right)}.
\end{aligned}$$

Problem 7**C&B Exercise 1.46, p. 43**

Seven balls are distributed randomly among seven cells. Let X_i be the number of cells containing exactly i balls. Find the pmf of X_3 .

Solution

Assume the seven distinguishable balls independently choose among the seven labeled cells with equal probability. There are $7^7 = 823,543$ equally likely assignments. Since three cells containing three balls would require nine balls, X_3 can take only the values 0, 1, 2.

Step 1: Count assignments with $X_3 = 2$. Choose the two triple-occupied cells in $\binom{7}{2}$ ways. In a fixed order of these two cells, choose their balls in $\binom{7}{3}\binom{4}{3}$ ways. The one remaining ball must go to one of the other five cells. Thus

$$N_2 = \binom{7}{2} \binom{7}{3} \binom{4}{3} \cdot 5 = 14,700.$$

Step 2: Count assignments with $X_3 = 1$. First mark a cell with exactly three balls, select its three balls, and place the other four balls in the other six cells. This counts

$$M = 7 \binom{7}{3} 6^4 = 317,520$$

pairs consisting of an assignment and a marked triple-occupied cell. Each assignment with $X_3 = 1$ occurs once, and each with $X_3 = 2$ occurs twice. Hence $M = N_1 + 2N_2$, and

$$N_1 = 317,520 - 2(14,700) = 288,120.$$

Step 3: Obtain the remaining probability by subtraction.

$$N_0 = 823,543 - 288,120 - 14,700 = 520,723.$$

Therefore the complete pmf is

$$\mathbb{P}(X_3 = x) = \begin{cases} 520723/823543, & x = 0, \\ 288120/823543, & x = 1, \\ 14700/823543, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Problem 8**C&B Exercise 1.47, p. 43**

Verify that each of the five functions in the exercise is a cdf.

Solution

A function F is a cdf if it is nondecreasing, right-continuous, and satisfies $F(-\infty) = 0$ and $F(\infty) = 1$.

(a) $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$, $x \in \mathbb{R}$. Since $\arctan x \rightarrow -\pi/2$ as $x \rightarrow -\infty$ and $\arctan x \rightarrow \pi/2$ as $x \rightarrow \infty$, the required limits are 0 and 1. Moreover,

$$F'(x) = \frac{1}{\pi(1+x^2)} > 0.$$

The function is continuous everywhere, so all three conditions hold.

(b) $F(x) = (1 + e^{-x})^{-1}$, $x \in \mathbb{R}$. As $x \rightarrow -\infty$, $e^{-x} \rightarrow \infty$ and $F(x) \rightarrow 0$; as $x \rightarrow \infty$, $e^{-x} \rightarrow 0$ and $F(x) \rightarrow 1$. Also,

$$F'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} > 0.$$

The function is continuous, hence is a cdf.

(c) $F(x) = e^{-e^{-x}}$, $x \in \mathbb{R}$. Here $F(x) \rightarrow 0$ as $x \rightarrow -\infty$, and $F(x) \rightarrow 1$ as $x \rightarrow \infty$. Continuity is immediate, and

$$F'(x) = e^{-x} e^{-e^{-x}} > 0,$$

so F is a cdf.

(d) The expression $1 - e^{-x}$ for $x > 0$ is completed to a function on $\mathbb{R}$ by setting

$$F(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-x}, & x > 0. \end{cases}$$

It has limits 0 and 1 at $-\infty$ and ∞ , respectively. It is constant on $(-\infty, 0]$ and increasing on $(0, \infty)$, since $F'(x) = e^{-x} > 0$ there. At zero, $\lim_{x \downarrow 0} (1 - e^{-x}) = 0 = F(0)$, so it is continuous there as well.

(e) Equation (1.5.6) on textbook p. 33 defines, for $0 < \epsilon < 1$,

$$F(x) = \begin{cases} \frac{1-\epsilon}{1+e^{-x}}, & x < 0, \\ \epsilon + \frac{1-\epsilon}{1+e^{-x}}, & x \geq 0. \end{cases}$$

The limits at $-\infty$ and ∞ are 0 and $\epsilon + (1 - \epsilon) = 1$. Both branches are increasing and continuous. At zero,

$$F(0-) = \frac{1-\epsilon}{2}, \quad F(0) = F(0+) = \frac{1+\epsilon}{2}.$$

The jump is upward with size ϵ , and the value at zero agrees with the right limit. Thus the function is globally nondecreasing and right-continuous, and it is a cdf with a point mass ϵ at zero.

Problem 9**C&B Exercise 1.49, p. 44**

Suppose $F_X(t) \leq F_Y(t)$ for every t , with strict inequality for at least one t . Prove the corresponding inequalities for the upper-tail probabilities of X and Y .

Solution

By the definition of a cdf and the complement rule,

$$\mathbb{P}(X > t) = 1 - F_X(t) \geq 1 - F_Y(t) = \mathbb{P}(Y > t) \quad \text{for every } t.$$

If t_0 satisfies $F_X(t_0) < F_Y(t_0)$, then

$$\mathbb{P}(X > t_0) = 1 - F_X(t_0) > 1 - F_Y(t_0) = \mathbb{P}(Y > t_0).$$

This proves both claims. No continuity or independence assumption is needed; the strict tail event $X > t$ is exactly the complement of $X \leq t$.

Problem 10**C&B Exercise 1.51, p. 44**

Of 30 microwave ovens, 5 are defective. Four are selected uniformly without replacement. Let X be the number of defective ovens selected. Find its pmf and cdf, and plot the cdf.

Solution

The pmf. There are $\binom{30}{4} = 27,405$ equally likely four-oven subsets. To select exactly k defective ovens, choose k of the 5 defective ovens and $4 - k$ of the 25 nondefective ovens. Thus

$$p_X(k) = \frac{\binom{5}{k} \binom{25}{4-k}}{\binom{30}{4}}, \quad k = 0, 1, 2, 3, 4,$$

and $p_X(x) = 0$ otherwise. This is a hypergeometric distribution. The following table lists exact and approximate values:

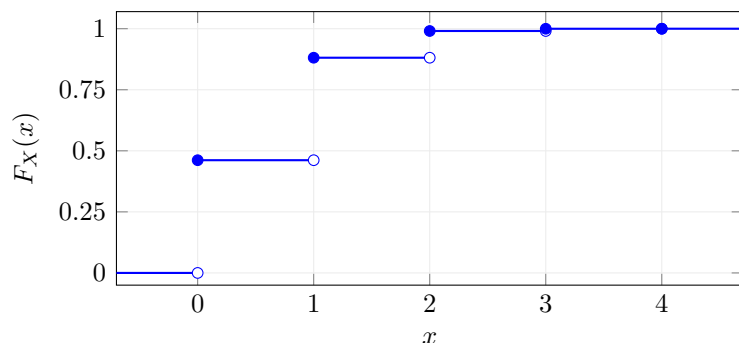
k	$27405 p_X(k)$	$p_X(k)$	$27405 F_X(k)$	$F_X(k)$
0	12650	0.461595	12650	0.461595
1	11500	0.419631	24150	0.881226
2	3000	0.109469	27150	0.990695
3	250	0.009122	27400	0.999818
4	5	0.000182	27405	1.000000

The cdf. Summing the pmf up to x gives

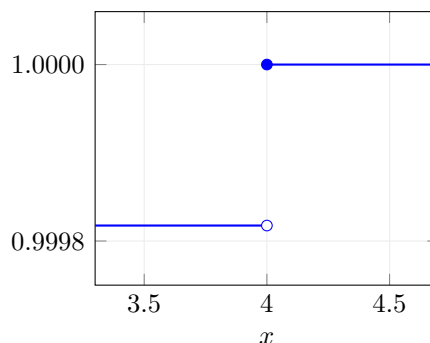
$$F_X(x) = \begin{cases} 0, & x < 0, \\ 12650/27405, & 0 \leq x < 1, \\ 24150/27405, & 1 \leq x < 2, \\ 27150/27405, & 2 \leq x < 3, \\ 27400/27405, & 3 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

Plot of the cdf. Each horizontal segment includes its left endpoint and excludes its right endpoint. Filled circles give the function values at the jumps; open circles give the left limits. The final jump at $x = 4$ has size $5/27405$ and is visible in the magnified panel.

Problem 10: CDF plot (Exercise 1.51)



Final jump (zoom)



Problem 11

C&B Exercise 1.53, p. 44

The annual high-water mark Y has distribution function $F_Y(y) = 1 - y^{-2}$ for $y \geq 1$. Verify that this is a cdf, find its pdf, and find the cdf of $Z = 10(Y - 1)$ after the change of origin and units.

Solution

(a) **Verify the cdf.** The full function is

$$F_Y(y) = \begin{cases} 0, & y < 1, \\ 1 - y^{-2}, & y \geq 1. \end{cases}$$

It has limits 0 and 1 at $-\infty$ and ∞ . It is constant below 1 and increasing above 1 because its derivative is $2y^{-3} > 0$ there. At the boundary, $F_Y(1) = 0$ and both one-sided limits equal 0. It is therefore continuous (in particular, right-continuous) everywhere and satisfies all cdf conditions.

(b) **Find the pdf.** Differentiation gives

$$f_Y(y) = \begin{cases} 2y^{-3}, & y > 1, \\ 0, & y \leq 1. \end{cases}$$

The choice of density value at $y = 1$ does not affect the distribution. As a check, $\int_1^\infty 2y^{-3} dy = 1$; there is no point mass at 1.

(c) **Transform the variable.** Since $Y \geq 1$ almost surely, $Z \geq 0$ almost surely. For $z \geq 0$,

$$\begin{aligned} F_Z(z) &= \mathbb{P}(10(Y - 1) \leq z) \\ &= \mathbb{P}\left(Y \leq 1 + \frac{z}{10}\right) = 1 - \left(1 + \frac{z}{10}\right)^{-2}. \end{aligned}$$

Thus

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ 1 - \frac{100}{(z + 10)^2}, & z \geq 0. \end{cases}$$

Problem 12**C&B Exercise 1.55, p. 44**

An electronic device has lifetime T with density $f_T(t) = \frac{1}{1.5}e^{-t/1.5}$ for $t > 0$. Its value is $V = 5$ if $T < 3$ and $V = 2T$ otherwise. Find the cdf of V .

Solution

The lifetime cdf is $F_T(t) = 1 - e^{-t/1.5}$ for $t \geq 0$. All lifetimes below 3 produce the same value 5, while lifetimes at least 3 produce values at least 6. Consequently V has a point mass at 5 and no possible values strictly between 5 and 6.

Case 1: $v < 5$. Neither branch can produce a value at most v , so $F_V(v) = 0$.

Case 2: $5 \leq v < 6$. Only the constant-value branch contributes:

$$F_V(v) = \mathbb{P}(T < 3) = 1 - e^{-3/1.5} = 1 - e^{-2}.$$

Case 3: $v \geq 6$. The two branches are disjoint, so

$$\begin{aligned} F_V(v) &= \mathbb{P}(T < 3) + \mathbb{P}(3 \leq T \leq v/2) \\ &= \mathbb{P}(T \leq v/2) = 1 - e^{-(v/2)/1.5} = 1 - e^{-v/3}. \end{aligned}$$

The required cdf is therefore

$$F_V(v) = \begin{cases} 0, & v < 5, \\ 1 - e^{-2}, & 5 \leq v < 6, \\ 1 - e^{-v/3}, & v \geq 6. \end{cases}$$

At $v = 5$ the jump is $\mathbb{P}(V = 5) = 1 - e^{-2}$. At $v = 6$ the two expressions agree, so there is no jump: $\mathbb{P}(V = 6) = \mathbb{P}(T = 3) = 0$.